Protein samples to be run on SDS-PAGE are added to sample solubilizing buffer containing beta mercaptoethanol (disrupt disulphide bridges), SDS, glycerol (to make the solution denser and enable proteins to sink in the gel) and bromophenol blue (tracking dye).

SDS -PAGE contain resolving gel, used for separation of proteins and stacking gels for concentrating the proteins prior to entry into resolving gel. Sodium dodecyl sulphate (SDS) is an anionic detergent, which binds to proteins, and provides a constant negative charge per unit mass. Protein-SDS complexes will therefore move towards the anode during electrophoresis and their mobilities are inversely proportional to the log of their molecular weights. Since the SDS impart proteins have the same charge per unit length, all proteins travel with the same mobility. However, as the mixture of proteins pass through the resolving gel, the proteins separate, owing to the molecular sieving properties of the gel. The smaller proteins move fast as they can pass through the pores of the gel. But larger proteins move slowly since they are retarded by frictional resistance due to sieving effect of the gels. When the dye reaches the bottom of the gel, the current is turned off. After electrophoresis, the gel is carefully removed from the glass plate, immersed in buffer and stained with appropriate stain solution (Fig.10.10)

Protein staining: Proteins can be detected using Coomassie Brilliant Blue G250 (CBB) solution. CBB dye stains protein with a detection limit of 40µg. For proteins of less quantity, another sensitive detection known as silver staining (1-5ng detection limit), can be performed.

Applications : SDS–PAGE is used to determine the molecular weight of proteins. To achieve this, a standard mixture of proteins of various molecular weight (molecular weight ladder) was added for direct comparison of migration distance. The molecular weights around 15-200kDa can be analyzed in this manner.

iii. Two -Dimensional gel electrophoresis

Two dimensional gel electrophoresis was introduced by O'Farell in the year 1975. It is a combination of two techniques, iso electric focusing and SDS-PAGE. Iso-electric focusing is an electrophoretic technique where proteins are separated based on their iso-electric point (pI). pI is the pH at which the amino acid does not migrate in an electric field (zwitterion form). When a gradient of pH is applied to a gel, and electric field applied to it, one end becomes more positive than the other. Relatively at all pH other than its iso electric point, proteins have a charge (positive or negative) and will be pulled to the opposite side of the gel. In two dimensional electrophoresis, proteins are separated based on isoelectric point and molecular mass. To accomplish this, proteins are first separated by isoelectric focusing where they are separated by their respective isoelectric point. Second dimension of separation is achieved through SDS-PAGE, where proteins are separated

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